

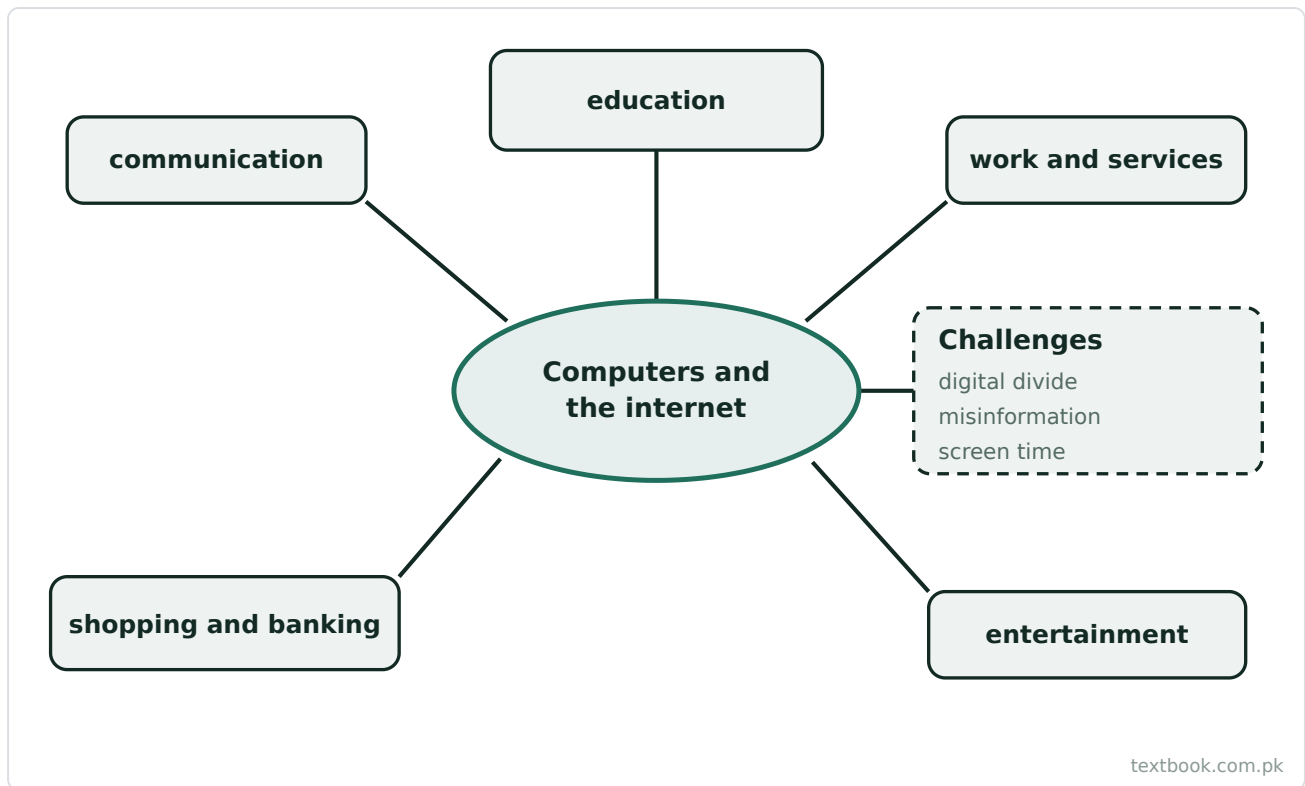
Draft - expert review pending

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Topic 1.4 - Computers, the internet, and society

A real classroom situation

Two classrooms, one hundred kilometres apart. In the first, a village primary school near Matiari gets its internet through a mobile signal on the head teacher's phone; when the signal holds, pupils watch a video of the sea for the first time, and their teacher downloads a lesson plan she could never have typed herself. In the second, a Karachi school's parents' group complains that pupils copy homework from the internet and stay up late on screens. Both stories are true, and both are about the same thing: computers and the internet have changed society, and the change cuts both ways. A teacher who understands both faces of the change can use the first and manage the second.



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Explanation

Impact of computers and the internet on society

The positive changes are easy to see. **Communication** became instant and almost free: a video call between Hyderabad and Toronto costs the same as a local message. **Education** opened up: a pupil in a government school can watch university lectures, and a teacher can join online training without travelling. **Work and services** moved onto screens: records that once filled rooms now fit in a database, and services from fee payment to passport tracking have online portals. **Commerce** grew new shapes, from city food-delivery apps to mobile wallets that let a shopkeeper in a small town accept digital payment without a bank branch.

The challenges are just as real. The **digital divide** is the gap between people who have good access to ICT and the skills to use it, and people who do not. Bourgeois and colleagues (2019) treat it as a first-order issue for information systems: the divide runs along income, geography, education, age and gender

lines, and it can widen existing inequalities instead of closing them. In Pakistan the divide is visible in the difference between a city household with broadband at home and a rural household where one shared phone is the family's only computer; most of the country's roughly 140 million broadband connections are mobile, not fixed (Pakistan Telecommunication Authority, 2025).

Misinformation travels the same wires as knowledge, and a reader who cannot judge a source inherits whoever shouted loudest. **Screen time** and online safety concerns follow pupils home, which is why Units 4 and 5 of this course exist.

A balanced view, then: computers and the internet are tools of enormous reach, and their impact depends on who can use them and how well they are taught to. That is exactly why a teacher's own computer literacy matters.

Applications of ICT in various fields

The second half of this topic is a tour of where ICT actually works in Pakistan. The table below summarises it; the paragraphs after give the flavour.

ICT across everyday fields in Pakistan

Field	What ICT does there	One Pakistan example
Education	Lessons, records and training	Online lecture platforms School record software
Health	Patient records, telemedicine and equipment	Video consultation services
Agriculture	Weather advisories and market prices	Advisory SMS services for growers
Banking	ATMs and mobile wallets	Branchless banking agents
Government	Online forms and identity records	e-filing and online passport tracking

ICT helps people access information, services and records.

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Education. Besides classroom use, ICT runs the system itself: examination boards process millions of papers and results, universities hold admission tests

online, and teacher training increasingly arrives as online courses. UNESCO's ICT Competency Framework for Teachers (2018) is built on exactly this expectation: a modern teacher moves from using technology, to deepening lessons with it, to helping pupils create with it.

Health. Hospitals keep digital patient records; laboratories report results electronically; telemedicine services connect patients in small towns to city specialists over a video call. The hybrid monitors from Topic 1.3 sit at the bedside end of this field.

Agriculture. Growers receive weather advisories and market price information as SMS messages, which shapes when to irrigate and when to sell. Mobile networks reach fields that extension officers cannot visit often.

Banking. ATMs, branch banking and mobile wallets together move money for people who never visit a branch. Branchless banking agents, often a single shopkeeper with a phone and a small machine, bring basic banking to villages.

Government. Identity records, tax e-filing, online passport tracking and government portals put services one screen away that once required queues and agents.

Activity: ICT in my field

Work in groups, one field per group. About 25 minutes.

1. Each group takes one field: education, health, agriculture, banking or government.
2. In your group, list three concrete applications of ICT in your field. For each, write one sentence on who uses it and what it replaces or improves.
3. Then find one Pakistan example you have seen yourself or heard about from family, and be ready to say where your information came from.
4. Jigsaw share: one member travels to each other group, learns their three applications in three minutes, and reports back. Assemble the full five-field picture in your notebook.

Check your understanding

1. Define the digital divide in one sentence. *(Remember)*
2. Give two positive impacts of the internet on society and one challenge, each in a single phrase. *(Understand)*
3. A school in a remote area has one shared smartphone with a weak signal and no computer lab. Name two ways the digital divide shows up in this school's daily learning, and one step the school could take anyway. *(Apply)*
4. Name three fields from this topic and one ICT application in each. *(Remember)*

Summary

Computers and the internet changed communication, education, work, commerce and services, and the same wires carry challenges: the digital divide, misinformation and screen-time concerns. ICT applications run through every major field in Pakistan, from SMS advisories for growers to mobile wallets and online government portals. For a teacher, the lesson is double: use the reach, and teach pupils to live with the risks.

Self-assessment checklist

Sign in to sync your ticks across devices.

- ☐ I can define the digital divide and give one example of it from Pakistan.
- ☐ I can name two positive impacts and two challenges of computers and the internet for society.
- ☐ I can name one ICT application in each of three different fields.

Try this at your practicum school

Interview one non-teaching staff member at your practicum school (a clerk, a lab attendant or the head teacher's office assistant) about how computers or phones have changed their work in the last ten years. Write four or five sentences: what they did before, what they do now, and one thing that got harder as well as one that got easier.

Summative task

Choose a school or workplace you know well. In one page, analyse the impact of ICT on it: describe two things that improved, one challenge that arrived with the technology, and one practical recommendation you would make to the head teacher or manager. Support each point with something you observed or a fact from this topic.

Marking guide (10 marks):

- Two improvements described concretely and connected to ICT: 3 marks.
- One challenge identified accurately (not a vague complaint): 2 marks.
- Recommendation is practical and follows from the evidence presented (Analyse): 3 marks.
- Writing is clear and within one page: 2 marks.

Further reading

- UNESCO. (2018). *UNESCO ICT Competency Framework for Teachers* (Version 3). <https://unesdoc.unesco.org/ark:/48223/pf0000265723>
- Bourgeois, D. T., Smith, J. L., Wang, S., & Mortati, J. (2019). *Information Systems for Business and Beyond* (2nd ed.), chapter 11. Open Textbook Library. <https://open.umn.edu/opentextbooks/textbooks/information-systems-for-business-and-beyond>
- Pakistan Telecommunication Authority. (2025). *Telecom indicators*. <https://pta.gov.pk/en/telecom-indicators>

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